HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Performed **microbiological testing** and **environmental monitoring** of manufacturing areas and cleanrooms, adhering to approved laboratory procedures and cGMP compliance.
- Collected and analysed samples from production utilities and critical control points, contributing to product quality assurance and patient safety initiatives.
- Reviewed laboratory data for accuracy and completeness, supporting the documentation process for deviations and out-of-specification (OOS) results.
- Collaborated with **Quality Assurance** and **Manufacturing** teams to address compliance gaps and implement continuous improvement strategies within the laboratory environment.
- Supported **microbiological validation** activities and method transfer protocols, ensuring laboratory readiness for new product introductions and process changes.
- Contributed to cross-functional investigations of OOT (out-of-trend) results, applying critical thinking to identify root causes and recommend corrective actions.
- Managed competing priorities within a laboratory setting, maintaining a strong focus on accuracy and compliance across multiple concurrent tasks and projects.

SKILLS

Laboratory Techniques – Microbiological Testing, Environmental Monitoring, Sample Collection, Data Review, cGMP Compliance, Laboratory Investigations, Method Transfer, Validation Activities

Quality Control – OOS/OOT Investigations, Deviation Management, Data Accuracy, Regulatory Compliance, Quality Assurance Collaboration

Tools & Technologies – Splunk, Dynatrace, AWS CloudWatch, Jira, Confluence, Linux Shell Scripting

Programming & Data – Python, SQL, Java Spring Boot, PostgreSQL, pandas, NumPy, scikit-learn

Data Visualization – Power BI, Tableau, DAX, Power Query

Project Management – Agile Methodologies, Scrum Ceremonies, Change Management, Cross-functional Teamwork

PROJECTS

Laboratory Data Integrity & Compliance Enhancement

- Developed Python scripts to automate the validation of **microbiological testing** data against defined quality parameters, reducing manual review time by **30%**.
- Designed and implemented interactive **Power BI dashboards** to visualize environmental monitoring trends and key performance indicators for the QC team, facilitating faster decision-making.
- Contributed to the documentation and review of standard operating procedures (SOPs) for laboratory processes, ensuring alignment with cGMP requirements and audit readiness.

Automated Reporting for Enterprise Systems

- Engineered Python ETL pipelines to extract and transform data from disparate enterprise systems, consolidating it into a central PostgreSQL data warehouse for operational reporting.
- Wrote complex **SQL queries** and performed data validation to ensure the accuracy and consistency of integrated data streams, supporting business analysts in generating critical reports.
- Managed scheduled data refresh jobs using cron on Linux servers, ensuring timely availability of data for downstream analytics and reporting tools.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)